

ALGEBRA BASICS



GRADE 6

WHAT IS A LINEAR EQUATION?

A linear equation is an algebraic number sentence that makes a straight line when you graph it.

Here's an example:

$$3x + 4 = 10$$





Basic Operations in Algebra

Addition, subtraction, multiplication and division are some of the operations used to create algebraic expressions.

Let's look at the following phrases and see how they can be translated into algebraic expressions.



Word Expression	Algebraic Expression	Simplified Expression
a increased by 5	$a + 5$	$a + 5$
2 less than a	$a - 2$	$a - 2$
a groups of b	$a \times b$	ab
a divided into b equal parts	$a \div b$	$\frac{a}{b}$



HOW DO WE SOLVE THEM?

Solving a linear equation means finding out what value the variable (letter) represents.

For example:

What does x represent in the equation $2x - 5 = 1$?





TO SOLVE LINEAR EQUATIONS, WE
CAN USE 'UNDOING STEPS'!

UNDOING STEPS = OPPOSITE OPERATIONS



LET'S TRY IT TOGETHER

Solve the linear equation: $2b = 18$

First step: Think about what $2b$ means.

$2b$ means $2 \times b$ (2 groups of b)

Second step: Think about what the opposite operation to multiplying is.

The opposite of multiplying is ...

DIVIDING!

So, we divide both sides by 2.

$$2b = 18$$

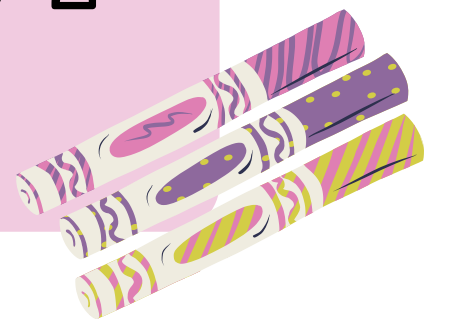
$$2b \div 2 = 18 \div 2$$

$$\cancel{2}b \div \cancel{2} = 18 \div 2$$

$$b = 9$$

Remember: We must divide BOTH sides of the equation by two to keep the equation balanced!

The multiply by 2 and divide by 2 cancel each other out.





ARE WE RIGHT?

We can use SUBSTITUTION to check if we have found the correct value for the unknown number in a linear equation.

Substitution in algebra involves swapping the variable (letter) for a particular number.

If the equation is BALANCED, we know we have found the correct value!

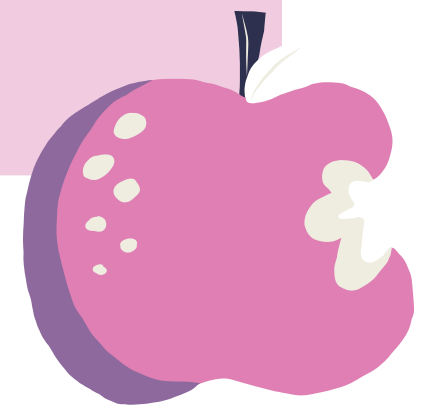
Let's substitute 9 into 'b'.

$$2b = 18$$

$$2 \times 9 = 18$$

$$18 = 18$$

The equation is balanced (both sides equal 18), so we know that $b = 9$ is the correct solution.



Let's solve this equation: $2x - 5 = 1$

$2x - 5$ means x is multiplied by 2, and then 5 is subtracted.

The opposite would be to
add 5 and divide by 2.



$$2x - 5 = 1$$

$$2x - 5 + \mathbf{5} = 1 + \mathbf{5}$$

First step: Add 5 to both sides.

$$2x - \cancel{5} + \cancel{5} = 1 + 5$$

$$2x = 6$$

$$2x \div \mathbf{2} = 6 \div \mathbf{2}$$

Second step: Divide both sides by 2.

$$\cancel{2}x \div \cancel{2} = 6 \div 2$$

$$x = 3$$



LET'S CHECK IT

Let's substitute 3 into 'x'.

$$2 \times 3 - 5 = 1$$

$$6 - 5 = 1$$

$$1 = 1$$

$x = 3$ is the
correct solution!

Like and Unlike Terms

Like Terms

Terms that have the same variable raised to the same power.

$$3x + 5x$$

$$7t + 6t$$

$$2xy + 6xy$$

$$5y^2 + 3y^2$$

Unlike Terms

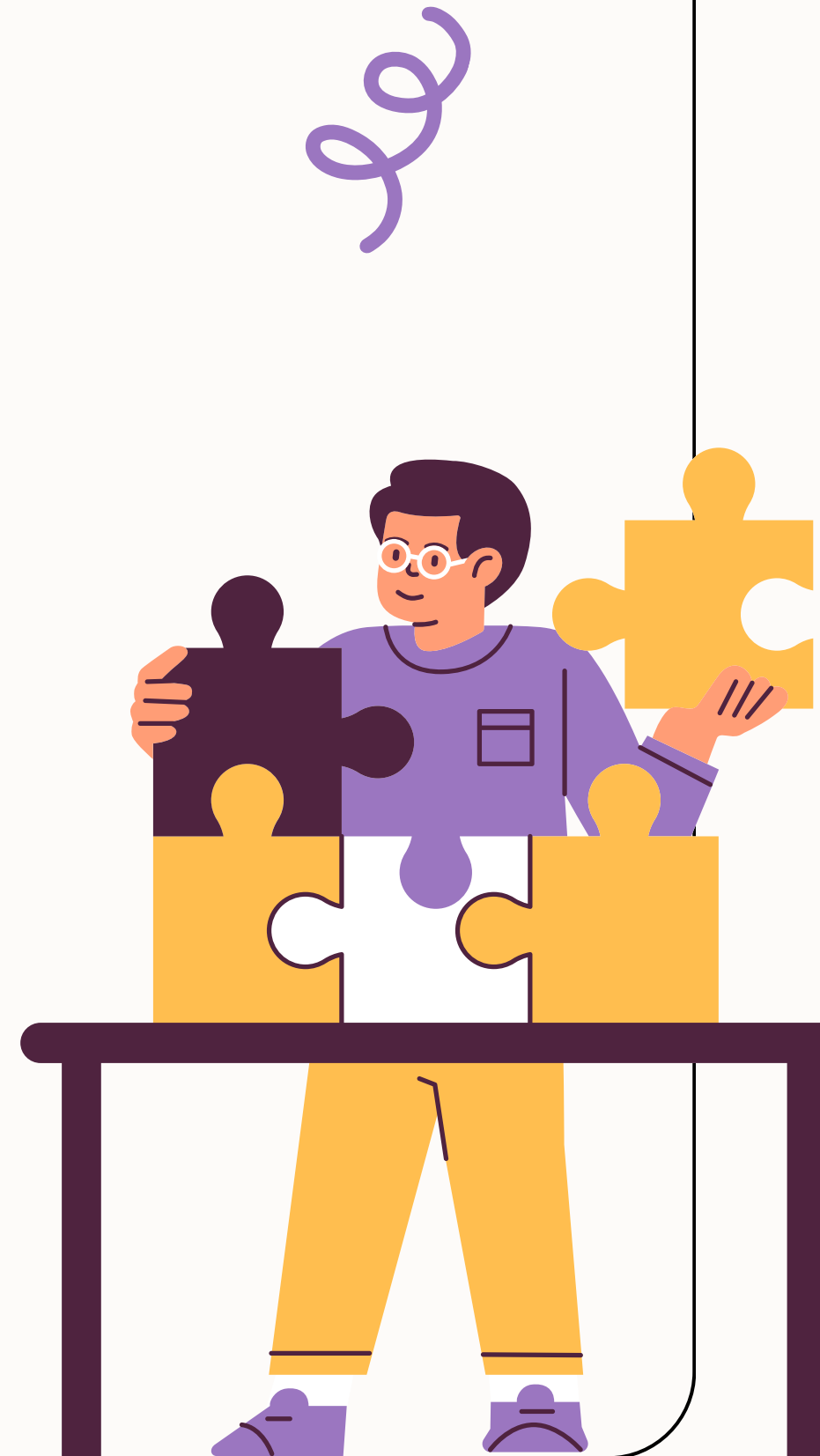
Terms that have different variables and/or different powers.

$$3x + 5y$$

$$7t + 6v$$

$$2xy + 6x$$

$$5y^2 + 3y$$



Tick the expressions with like terms

$$4x + 5x$$

$$4xy + 5xy$$

$$4x + 5y$$

$$4x^2 + 5x$$

$$4xy + 5y$$

$$4y + 5y$$

Tick the expressions with like terms

$$4x + 5x$$



$$4xy + 5xy$$



$$4x + 5y$$

$$4x^2 + 5x$$

$$4xy + 5y$$

$$4y + 5y$$



Example

$$7x + 4y - 3x + 2y$$

- ① Identify like terms

$$7x + 4y - 3x + 2y$$

- ② Group like terms

$$7x - 3x + 4y + 2y$$

- ③ Combine like terms

$$4x + 6y$$



Checkpoint

Match up the expressions with their simplified answer



$$9y - 7x + 3y + 4x$$

$$x + 2y$$

$$4x + 3y + 5x - 5y$$

$$12y - 3x$$

$$5y - 7x - 3y + 8x$$

$$9x - 2y$$

Answer Key

How many did you get correct?



$$9y - 7x + 3y + 4x$$



$$12y - 3x$$

$$4x + 3y + 5x - 5y$$



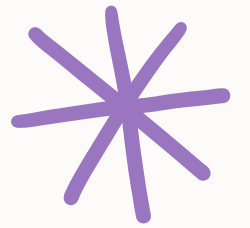
$$9x - 2y$$

$$5y - 7x - 3y + 8x$$



$$x + 2y$$

Challenge

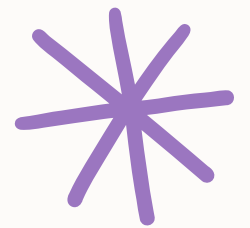


Can you simplify the following expression?

$$6x - 3x^2 + 4xy + 7x^2 - 5xy - 4x$$



Solution



- ① Identify like terms

$$6x - 3x^2 + 4xy + 7x^2 - 5xy - 4x$$

- ② Group like terms

$$6x - 4x + 4xy - 5xy + 7x^2 - 3x^2$$

- ③ Combine like terms

$$2x - xy + 4x^2$$



Remember these steps:

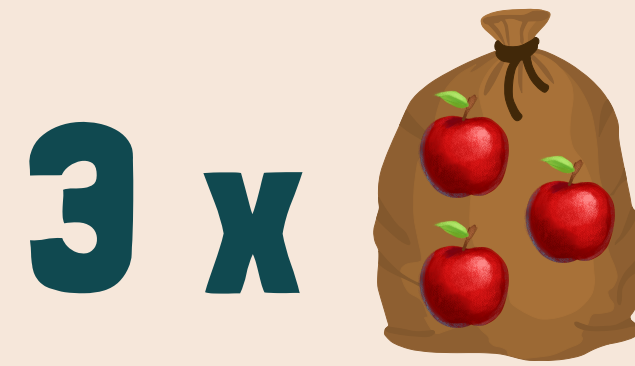
- ① Identify like terms
- ② Group like terms
- ③ Combine like terms



Well Done!

Simplifying Terms

$y + y + y$ is the same as 3 multiplied by y .

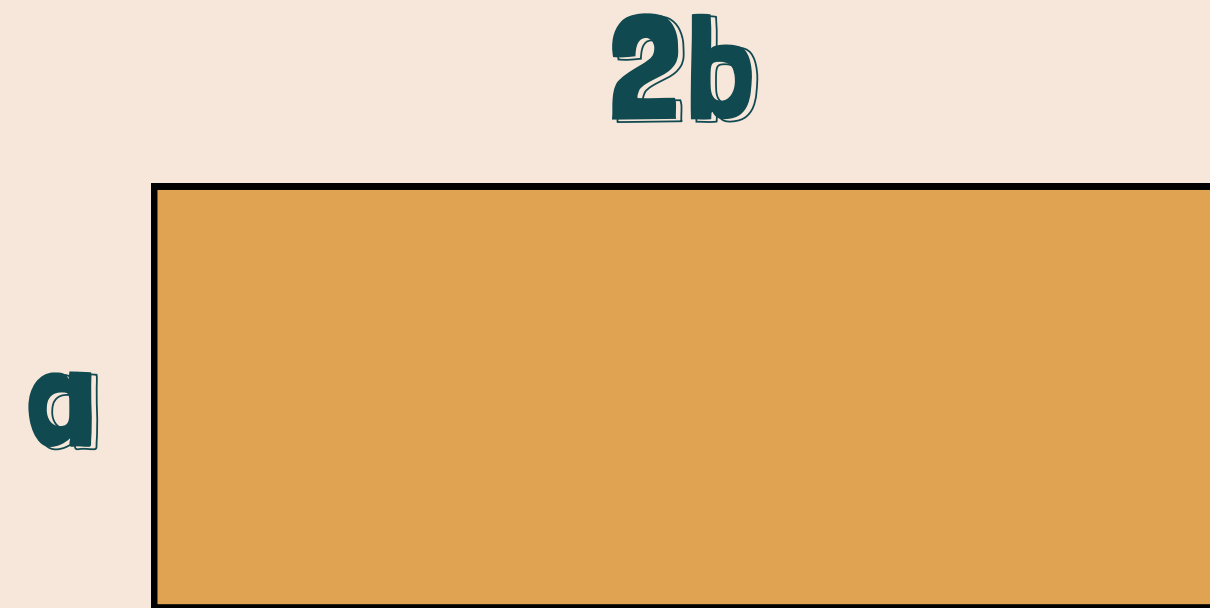


$$y + y + y$$

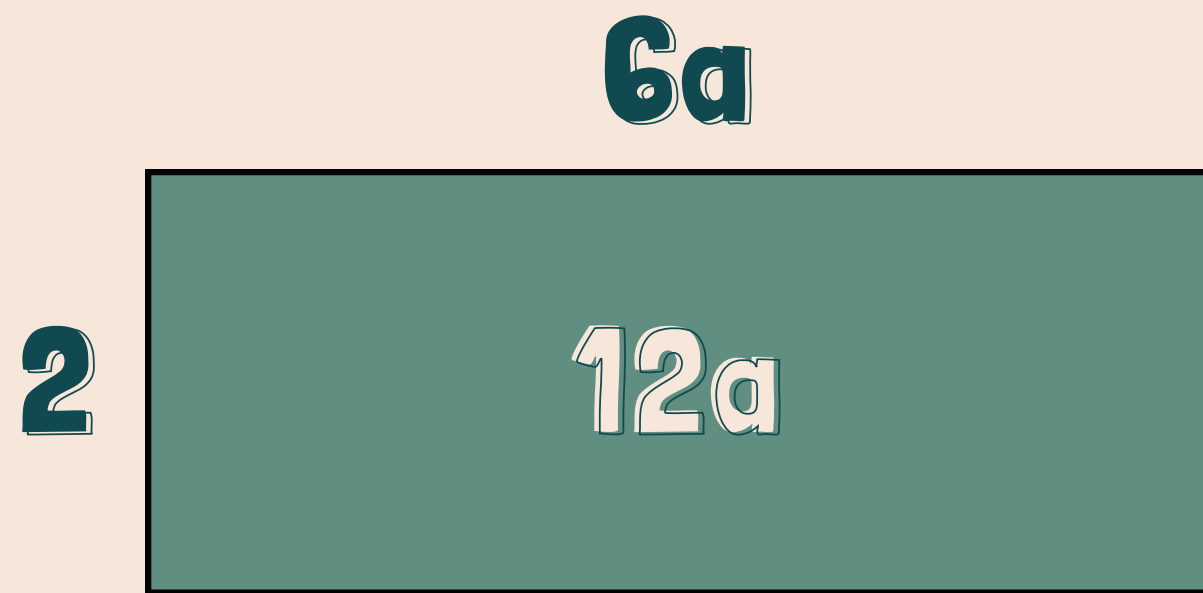
$$3 \times y$$

Multiplying Terms using the Area Model

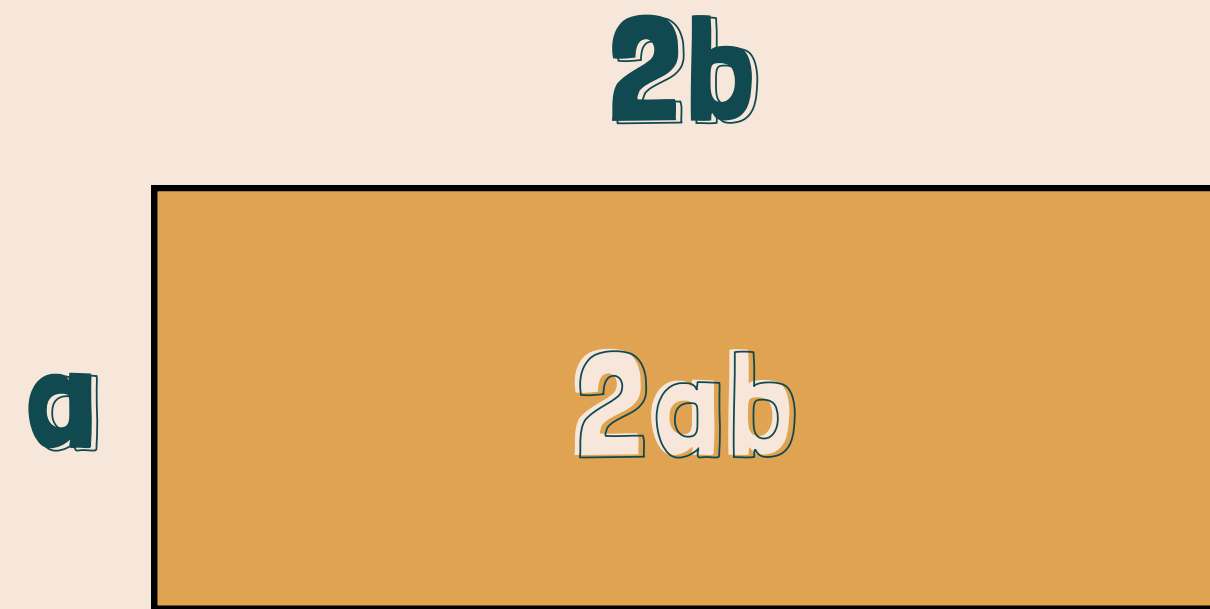
Can you figure out an expression for the area of each rectangle?



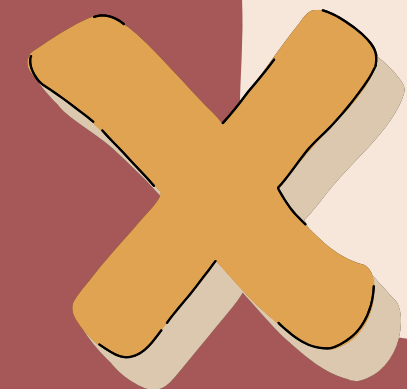
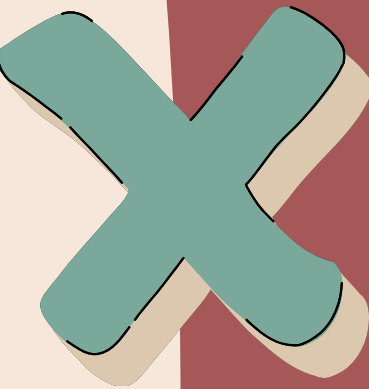
Multiplying Terms using the Area Model



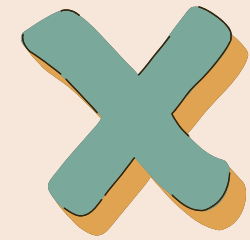
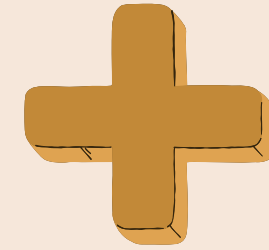
$$2 \times 6a = 12a$$



$$a \times 2b = 2ab$$



More Examples



* $3a \times 5 = 15a$

* $2a \times 3b = 6ab$

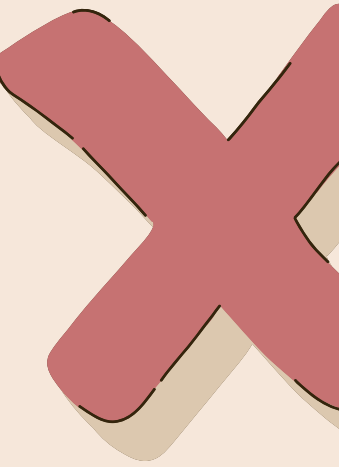
* $3a \times 4a = 12a^2$

Key Points

In algebra, $a \times b$ can be written as 'ab'.

When multiplying terms, you need to multiply the numbers together and then multiply the letters separately.

Have a go...



$$2a \times 3$$

$$4a \times 6b$$

$$6a \times b$$

$$3a \times a$$

$$7a \times 4b$$

$$7a \times 2a$$

Answer Key

$$2a \times 3 = 6a$$

$$4a \times 6b = 24ab$$

$$6a \times b = 6ab$$

$$3a \times a = 3a^2$$

$$7a \times 4b = 28ab$$

$$7a \times 2a = 14a^2$$

Multiplying Terms with Indices

$$2a^2b \times 3ab$$

$$\boxed{2} \times \boxed{a} \times \boxed{a} \times \boxed{b} \times \boxed{3} \times \boxed{a} \times \boxed{b}$$

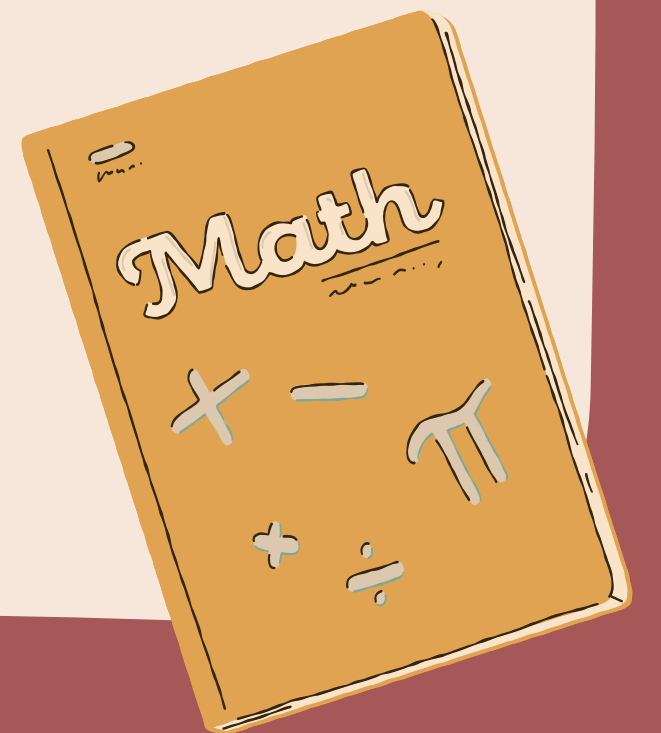
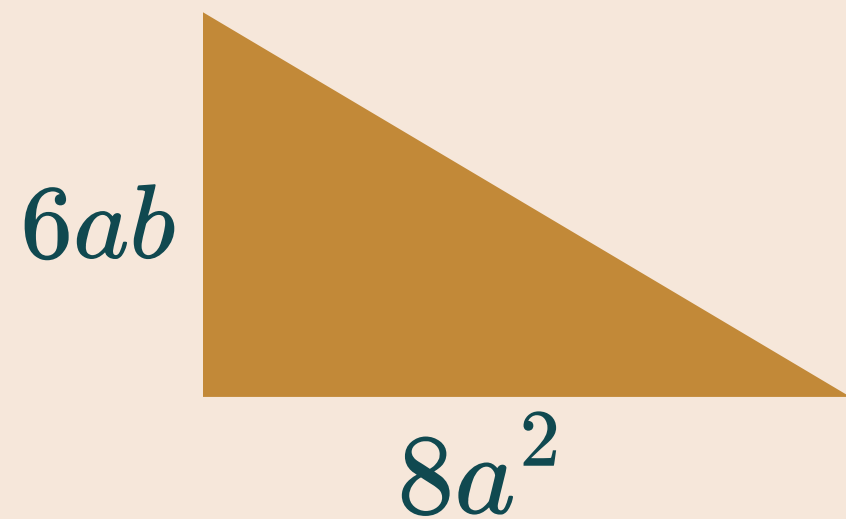
$$= \boxed{6} \times \boxed{a^3} \times \boxed{b^2}$$

$$= 6a^3b^2$$

Ready for a challenge?

1. Simplify the expression $4ab^2 \times 0.5a^2$

2. Find an expression for the area of the triangle

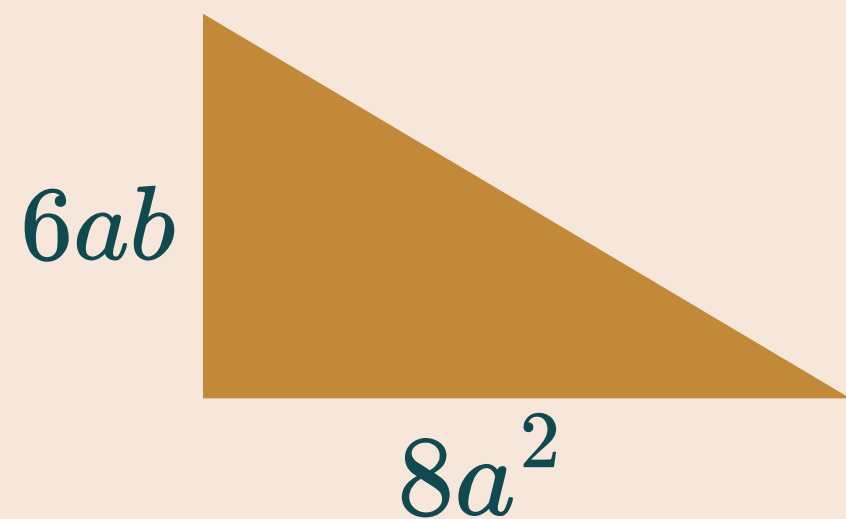


Solutions

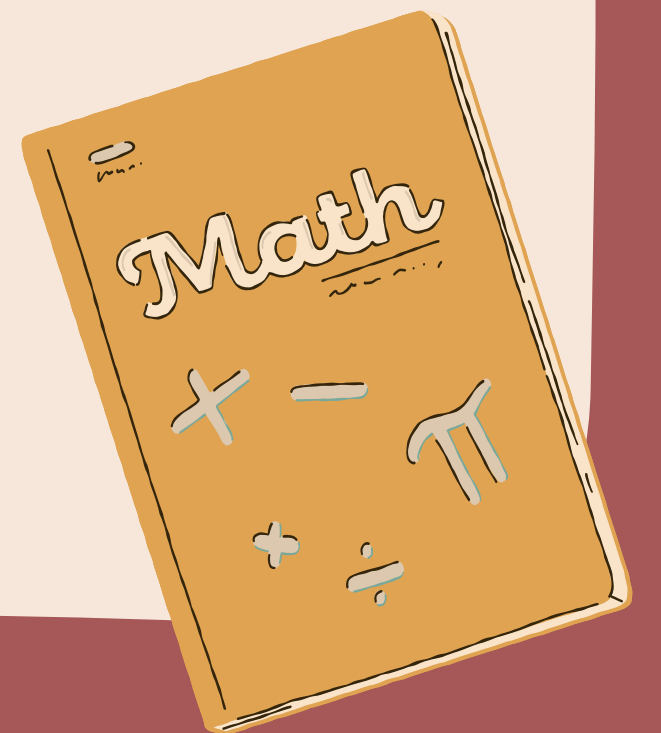
1. Simplify the expression $4ab^2 \times 0.5a^2$

$$= 2a^3b^2$$

2. Find an expression for the area of the triangle



$$\begin{aligned} \text{Area} &= \frac{1}{2} \times 6ab \times 8a^2 \\ \text{Area} &= 24a^3b \end{aligned}$$





The Notation

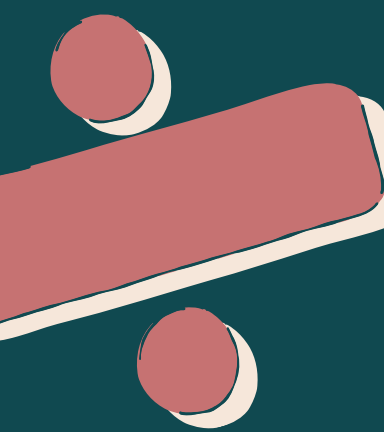
$$a \div 3$$

$$\equiv \frac{a}{3}$$

In algebra, division is represented by writing the terms as a fraction.

$a \div 3$ is written

as $\frac{a}{3}$.



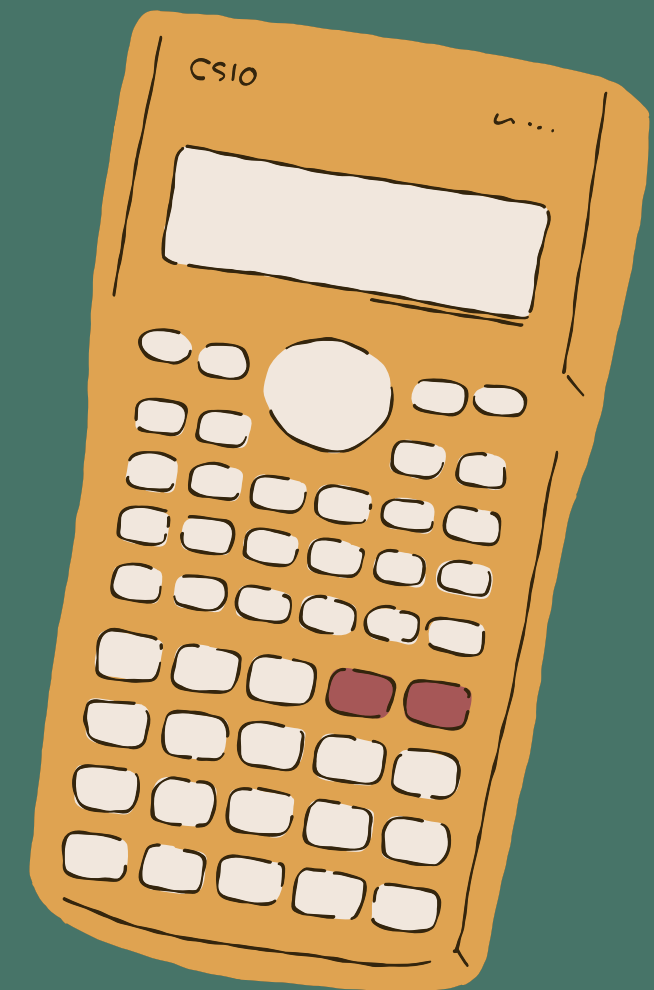
Example

$$15a \div 5$$

$$= \frac{15a}{5}$$

$$= 3a$$

Rewriting a question as a fraction allows you to identify common factors that can be simplified.

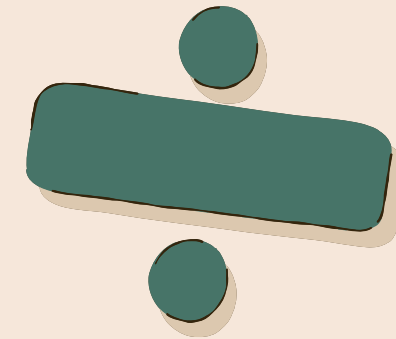
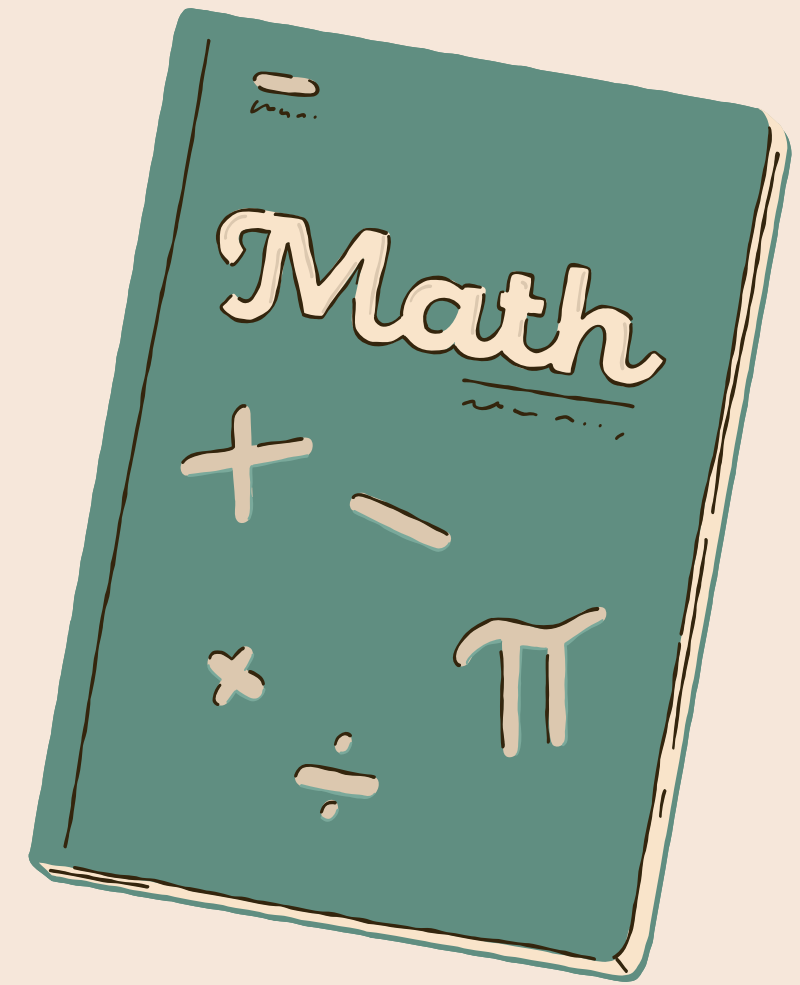


Quickfire Questions

1. $14x \div 2$

2. $15b \div b$

3. $27xy \div 3x$

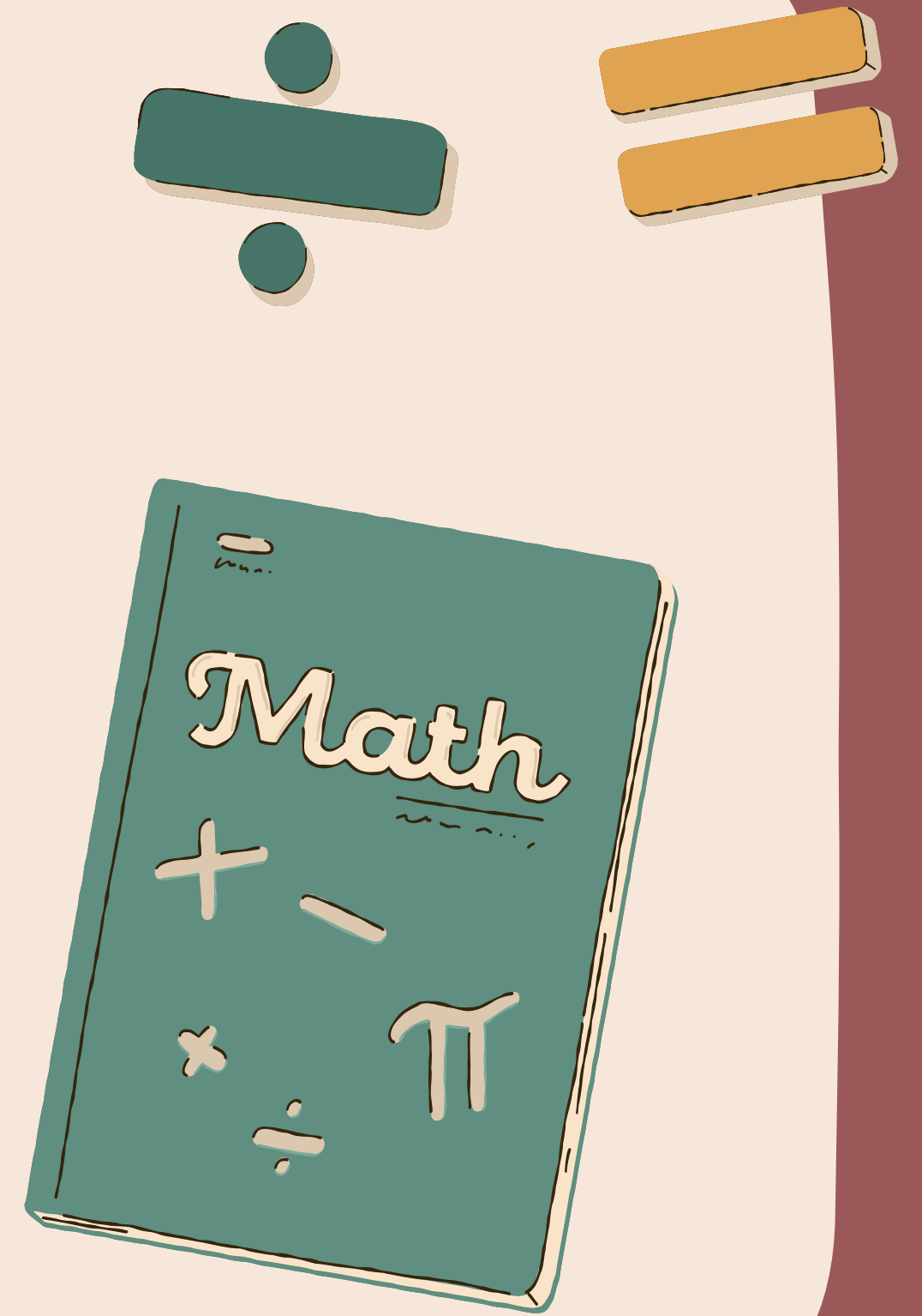


Answers

1. $14x \div 2$ $7x$

2. $15b \div b$ 15

3. $27xy \div 3x$ $9y$



1. Simplify the numbers.

$$\frac{24}{18} = \frac{4}{3}$$

Diagram showing the simplification of $\frac{24}{18}$ to $\frac{4}{3}$ by dividing both numerator and denominator by 6. Arrows indicate the division process: $\frac{24}{18} \xrightarrow{\div 6} \frac{4}{3}$.

2. Cancel out any common terms.

y can be cancelled out.

Harder Example

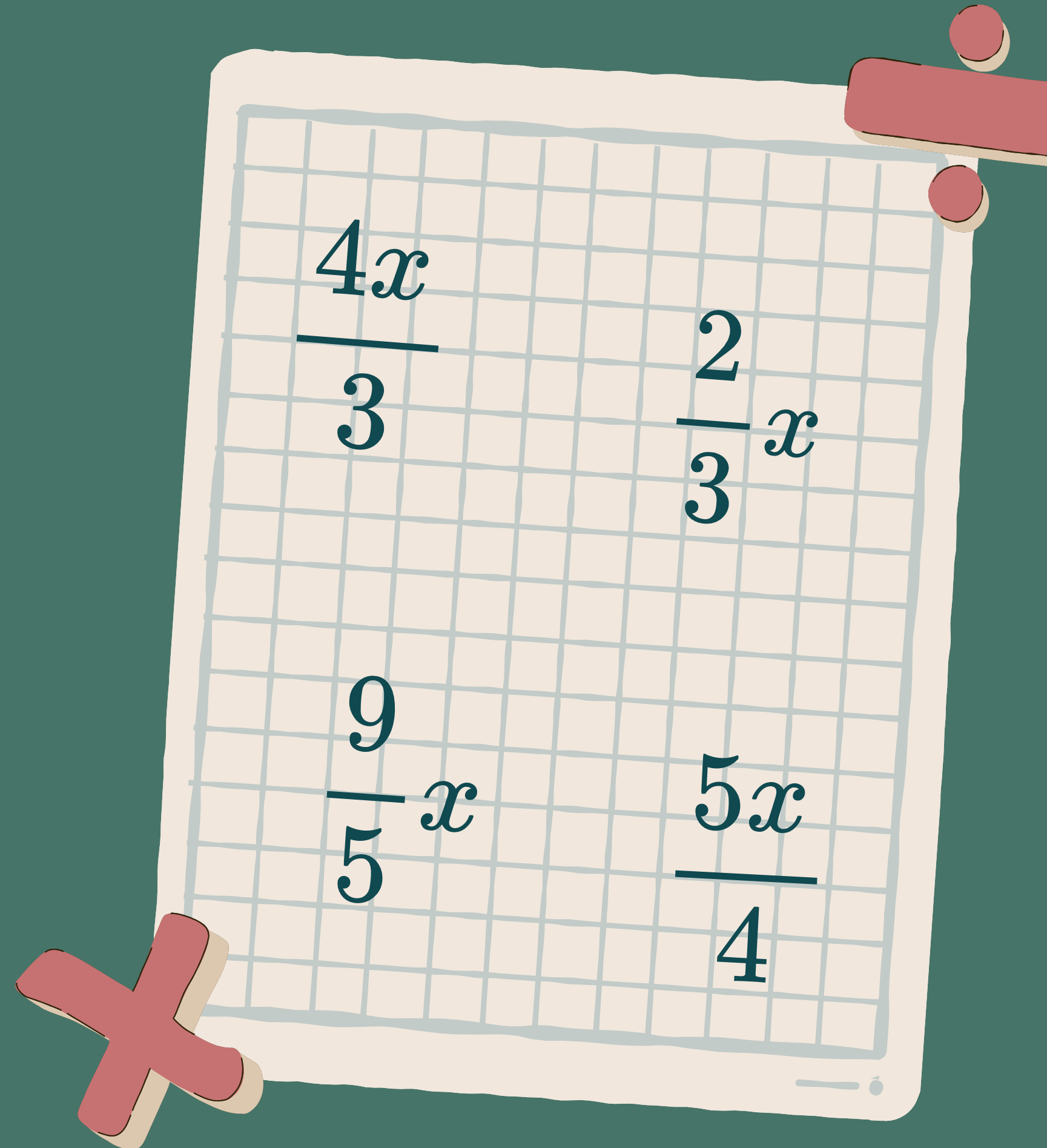
$$\frac{24xy}{18y}$$

$$= \frac{4x}{3}$$

Coefficients

A coefficient is a number that is multiplied by the variable.

In algebra, coefficients are written as fractions, rather than decimals as they are more accurate and easier to manipulate and simplify. Using decimals can lead to rounding errors.



Practice Questions

1. $\frac{15x}{6}$

2. $\frac{24y}{8x}$

3. $\frac{40xy}{6y}$

4. $\frac{28y}{42xy}$

Answer Key

1. $\frac{15x}{6}$

2. $\frac{24y}{8x}$

3. $\frac{40xy}{6y}$

4. $\frac{28y}{42xy}$

$$\frac{5x}{2}$$

$$\frac{3y}{x}$$

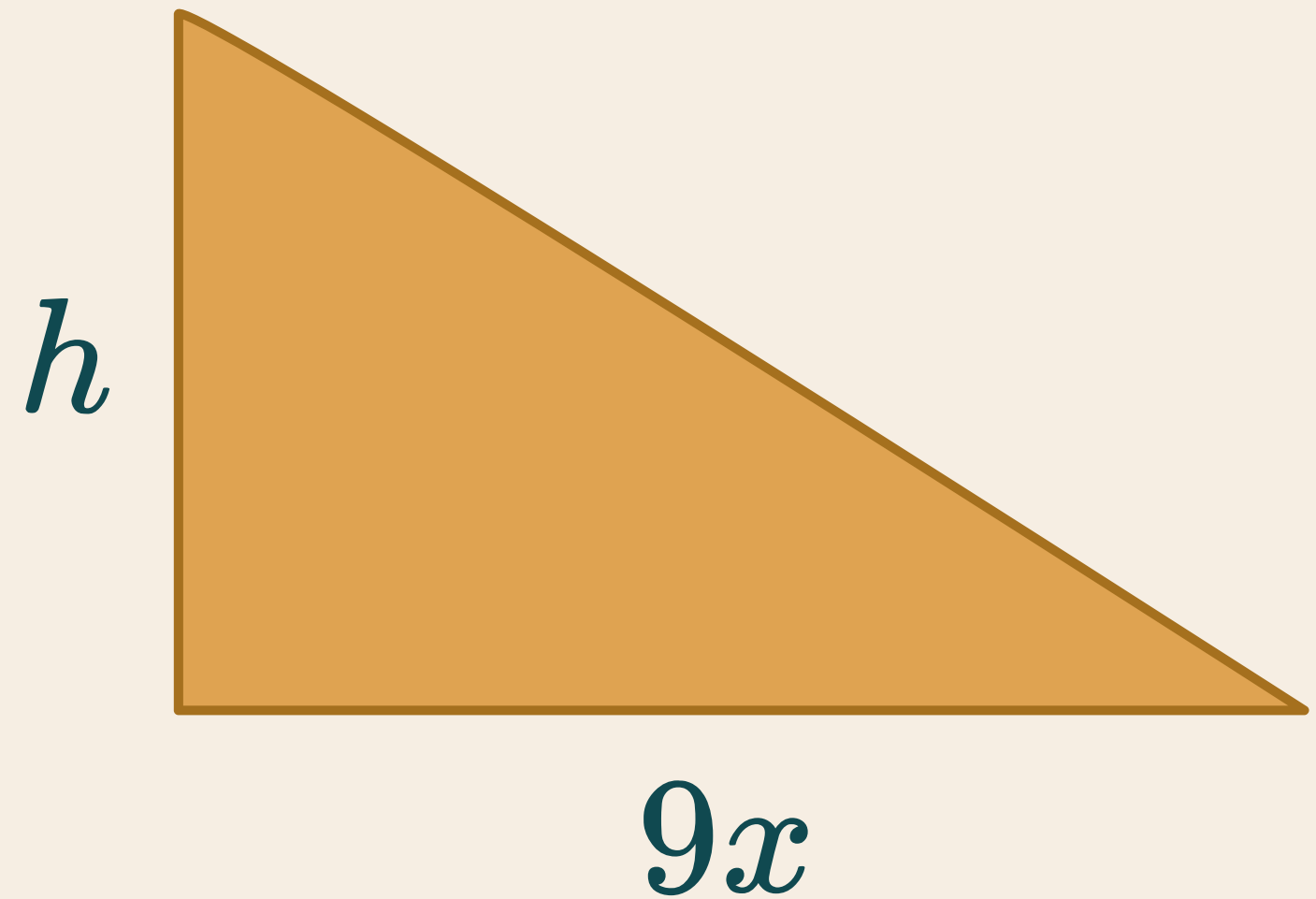
$$\frac{20x}{3}$$

$$\frac{2}{3x}$$

Challenge

Find an expression for the height of the triangle, given the area:

$$Area = \frac{27xy}{2}$$

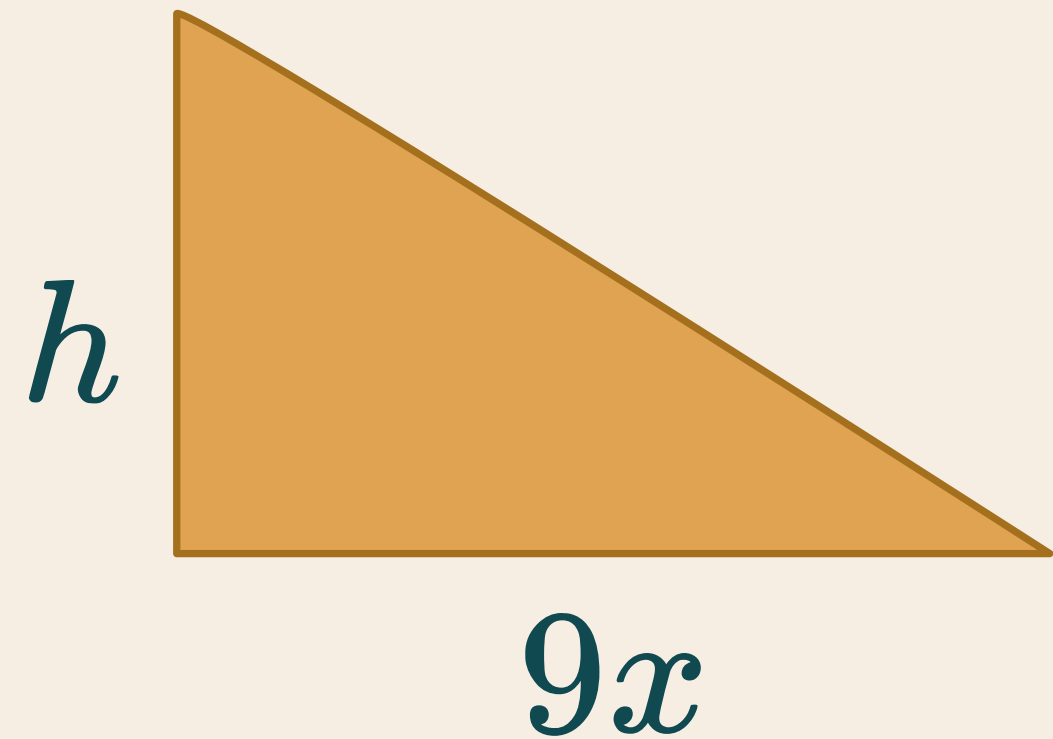


Answer Key

Find an expression for the height of the triangle, given the area:

$$h = 3y$$

$$Area = \frac{27xy}{2}$$



$$\frac{1}{2} \times 9x \times h = \frac{27xy}{2}$$



THANK YOU